

PUMP-1 Fluid Pump

Food-safe automated fluid pump for Home Assistant watering and refills

The Apollo PUMP-1 is a compact, food-safe fluid pump that integrates with ESPHome and Home Assistant to automate watering and refills. It pairs seamlessly with the PLT-1 plant sensor for hands-off plant care, and includes two ports for capacitive fluid sensors that detect water presence — so the pump stops automatically when watering is complete or a reservoir runs empty, protecting your system.

Beyond plants, the PUMP-1 handles a range of jobs: filling fish tanks, refilling robot-vacuum mop tanks and pet bowls, topping up coffee makers, and even acting as a drain pump. It runs on USB-C, connects over Wi-Fi, and is fully local and open-source.

Key Specifications

Product	PUMP-1 Fluid Pump
Manufacturer	Apollo Automation (Versailles, KY, USA)
Type	Mini DC fluid pump
Microcontroller	ESP32-C6-Mini
Firmware	ESPHome — open source
Connectivity	Wi-Fi (2.4 GHz)
Power input	5 V USB-C
Flow rate	Up to 900 ml/min
Pump head	14 ft+ vertical, 33 ft horizontal (tested)
Fluid sensing	Two capacitive fluid-level sensor ports
Dimensions	86.4 x 33 x 45.7 mm (3.4 x 1.3 x 1.8 in)
Material	Food-safe for liquids
In the box	PUMP-1; Apollo sticker

Configurations & Pricing

Item	Price (USD)
PUMP-1 Fluid Pump	\$32.99

Add-ons available separately: replacement tubing, fluid sensor, and 1 L bottle. Designed to pair with the Apollo PLT-1 Plant Sensor.

Features & Use Cases

- **Automated plant watering:** pair with the PLT-1 for a fully automated, sensor-driven watering system.
- **Fish tank fill & maintenance:** automate aquarium top-offs and water changes for a stable environment.
- **Robot-vac mop refills:** automatically refill the mop tank so cleaning never runs dry.
- **Pet bowl refills:** keep fresh water topped up throughout the day.
- **Coffee maker supply:** manage water levels for smarter morning routines.
- **Drain pump:** remove liquid from tanks, trays, or containers for cleaning and maintenance.

Important Notes

- **Dry-run protection:** the two capacitive fluid sensors let the pump stop automatically when watering completes or a reservoir empties.
- **Food-safe materials:** wetted parts are food-safe, suiting it to drinking water and food-adjacent uses.
- **Sized for the task:** flow is well suited to single plants and small reservoirs; size accordingly for larger or multi-pot setups.

Documentation & Resources

- **Product page:** apolloautomation.com/products/pump-1-fluid-pump
- **Wiki / setup guide:** wiki.apolloautomation.com/products/pump1
- **Pairs with:** apolloautomation.com/products/plt-1
- **Source & CAD files:** github.com/ApolloAutomation
- **Community:** [Apollo Automation Discord](#)